

Exploring Innovative Machine Learning Models for Soil CO₂ Emission Forecasting

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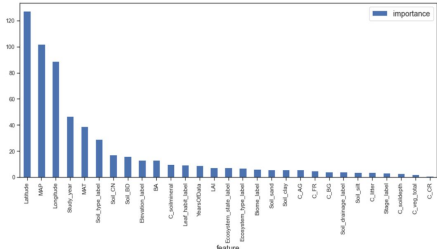
Q3: Data Analysis and Results

Model Performance Comparison

Evaluation metrics	RF	GB	XGBoost	DNN
MAE	186.97	206.63	179.43	331.08
RMSE	311.43	324.87	297.90	500.71
R ²	0.700	0.673	0.726	0.225

XGBoost model has the best performance in terms of predictive accuracy and error reduction compared with the other three models:

- Lowest MAE and RMSE
- Highest R²



SHAP analysis reveals that Mean Annual Temperature (MAT) and Mean Annual Precipitation (MAP) are the most influential environmental factors, alongside geographic features such as latitude and longitude, aligning with previous studies.

Q1: Research Question

- How can AI/machine learning framework help us accurately predict CO₂ emissions from soil respiration based on spatial, temporal, climatic, and other observable variables?
- Can AI/machine learning help us understand the environmental drivers behind soil respiration process?

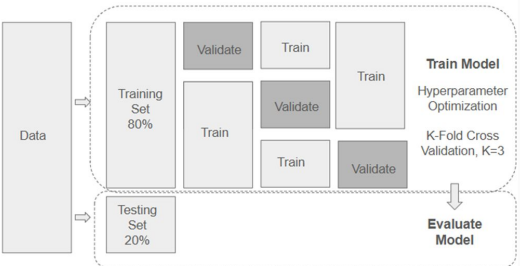
Q2: Methodology

Model selection:

Proposed Model: XGBoost;
Baseline models: Random Forest (RF),
Gradient Boosting (GB)

Hyperparameter optimization:

GridSearchCV with K-fold cross-validation.



Q4: Interpretation and Conclusions

- An innovative ensemble model, XGBoost, is developed to predict annual Rs from spatial, temporal, climate, and other independent variables. The XGBoost model delivers the best performance, achieving a 4% reduction in MAE and RMSE and a 3.7% improvement in R² compared to the Random Forest model, previously identified as the top-performing approach.
- SHAP (SHapley Additive exPlanations) is employed as a robust method for model explainability, enabling the assessment of feature impacts on predictions both globally and at the individual observation level.

